Evaluation of Family Planning Policy in India Using Synthetic Control Methods

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Abstract

Publicly funded family planning services have a considerable influence on the fertility decisions of women and, consequently maternal and child health. In November, 2016 an expansive family planning program was launched in 145 districts in seven provinces of India that had previously shown high fertility rates. The program featured (i) increase access to a range of contraceptives, (ii) monetary incentives for beneficiaries who adopt family planning methods, and (iii) information dissemination using mobile vans covering all blocks of a district. This paper documents the effects of this family planning program on a broad set of maternal and child outcomes. I use the generalized synthetic control method to estimate the average treatment effect on the treated districts utilizing administrative data from Health Management Information System of India aggregated to the district level for periods 2009-19. The estimates suggest that the program increased deliveries by 4.6, institutional deliveries of infants by 3.9 and infants weighed at birth by 3.2 per one thousand women of reproductive age. New cases of pregnant women with hypertension decreased by 2 per one thousand women of reproductive age. Heterogeneous treatment effects suggest that the program decreased the fraction of infants born with low-birth-weight in districts with high initial economic activity and increased it in districts with low initial economic activity.

1 Introduction

Increasing access to modern contraceptive methods has been shown to enable women and households to have more control over fertility decisions and other life choices such as labour force participation and hours worked (see Bailey (2006) Bailey (2010); Lindo and Packham (2017), Ananat and Hungerman (2012); Bailey et al. (2012); Goldin and Katz (2000), Guldi (2008)). These modern contraceptives can be short-term methods such as male and female condoms, emergency pills, over-the-counter (OTC) pills, or long-acting reversible contraceptive (LARCs) methods such as intrauterine devices, injectables, and other subdermal implants. The use of LARCs in particular has proven to be effective in preventing unwanted pregnancies, especially among young women (99.9% versus 82% for condoms) when compared to other traditional modern contraceptives (Group (2018)). However, the uptake of LARCs can be low among younger women due to high up-front costs, lack of access, misinformation, literacy or provider reluctance (Romero et al. (2015)). Short-acting contraceptives too can be subject to behavioural biases (Action (2015)), for example, a strong present bias may lead to discontinuation of the pill or inattention could lead to forgetting to take the pill in the prescribed intervals of time.

In the context of developing countries, these shortcomings in the efficacy of and access to contraceptives are only heightened given the lack of health infrastructure, urbanization and literacy. Nearly half of the women 20–24 years of age in Asia and Africa are married by age 18, putting them at an increased risk of early pregnancy, maternal disability, and death. Furthermore, half of all adolescent births occur in just seven countries: Bangladesh, Brazil, Democratic Republic of the Congo, Ethiopia, India, Nigeria, and the USA (Bhutta et al. (2014)). Among the SAARC (South Asian Association for Regional Cooperation) countries, adolescents in India comprise one-fifth of the total population in the region (Mehra and Agrawal (2004)).

This paper estimates the impacts of a contraceptive policy implemented in India in 2016 known as the **"Mission Parivar Vikas"** (MPV) program. The MPV program aimed to increase contraceptives and family planning services in 145 high fertility districts in seven states of the country. These districts contribute to approximately 28 percent of India's population, at least 30 percent of maternal deaths and 50 percent of infant deaths (Health and Welfare (2016)). I use administrative data aggregated at the district level acquired from the Health Management Information System (HMIS) for the periods 2009-19 to estimate the treatment effect of the policy on maternal and child. My analysis uses the generalized synthetic control (GSCM) method proposed by Xu (2017). This method uses a data-driven approach to weight untreated districts in the donor pool to most closely match the average outcomes in the treated districts. This method extends the synthetic control method (SCM) proposed by Abadie et al. (2010) by allowing for multiple treated units and variable treatment periods.

My results suggest that on average the MPV program led to an increase in deliveries by 4.6 deliveries per one thousand women of reproductive age in treated districts. It increased births in health institutes

and weighed infants by 3.9 and 3.2 per one thousand women of reproductive age respectively. Moreover, the *MPV* program reduced new cases of hypertension in pregnant women by 1.9 per one thousand women of reproductive age. Findings on heterogeneous treatment effects conditional on initial health outcomes and levels of economic activity suggest that the *MPV* program decreased fraction of infants with birthweight below 2.5 kilograms by 5.4% in districts with higher initial economic activity and increased it by 2.8% in districts that have low initial economic activity.

The remainder of the paper is organized as follows. Section 2 summarizes the literature on family planning policies and discusses on the contributions of this paper relative to previous work. Section 3 provides background information on the institutional setting of the *MPV* program. Section 4 describes the data used in this study and presents the empirical methodology. Section 5 presents the main results. Section 6 presents documentation of heterogeneous treatment effects of the *MPV* program and the mechanism driving the results. Section 7 concludes.

2 Literature Review

Existing work on the impacts of family planning services highlights the role of mediators such as preferences of consumers, market forces and state interventions in the presence of existing contraceptives and policies in explaining take-up and health outcomes. While clinical trials have proven the effectiveness of emergency contraceptive (EC) pills (Glasier and Baird (1998); Glasier et al. (2004); Raymond et al. (2004); Raine et al. (2005)), only a limited set of studies investigate the public health benefits from such policy e.g., reduced unintended pregnancy or abortion rates by improved access to EC pills. Clinical trials also rely on small-scale experimental data or quasi-experiments in specific areas within the country that may not be appropriate to estimate the potential social benefit a large-scale policy or health intervention produces. Pfeifer and Stockburger (2023) use a difference-in-differences methodology to estimate the impact of a switch from prescription-only EC pills to OTC pills in Europe in 2015. They find a substantial increase in sales and manufacturer's revenue but no effect on abortions, consistent with other contemporary studies (Durrance (2013); Girma and Paton (2006), Girma and Paton (2011); Gross et al. (2014); Mulligan (2016)). They also find that the policy switch from prescription-only EC pills to OTC pills increased fertility by 4% among women aged 25-34. The authors interpret their results as a decrease in birth control pill (despite the higher sales of the easy to access OTC pills) driving the increase in fertility. Other studies find that OTC access increases unprotected sex, the transmission of sexually transmitted diseases, unplanned pregnancies and abortions (Gross et al. (2014); Durrance (2013); Mulligan (2016)).

A number of other studies estimate the effects of increasing the supply and access to contraceptives in both developed (Luca et al. (2021)) and developing countries (Herrera-Almanza and Rosales-Rueda

(2020); Anukriti et al. (2022)). An important theme from these studies is that both monetary and non-monetary costs that women face in accessing health services is important for determining take-up and health outcomes, especially in developing countries.

A large literature documents the impact on in-utero conditions and finds that adverse conditions leads to negative health outcomes for infants that persist into adulthood (Currie and Almond (2011); Black et al. (2007); Hanson and Gluckman (2015); Oreopoulos et al. (2005); Conley et al. (2003)). Another groups of studies examines the impact of other policies such as cash transfers, increasing the availability of contraceptives and improving public health and family planning treatments on fertility and infant health (Falcão et al. (2023); Hatch et al. (2015); Oster (2006); Packham (2017)). This work documents the role of multifaceted policies and public health investments on the outcomes of mothers and infants.

This paper investigates the impact of the understudied *MPV* program in India on maternal and child outcomes. Agarwal et al. (2023) use two waves of survey data from the National Family Health Survey(NFHS) for the years 2015-16 and 2019-21 and a difference-in-differences (DiD) methodology to estimate the impact of the *MPV* program on fertility. They find decline in number of births 0.022 units more in *MPV* districts than that in control districts. My work builds on and aims to improve upon this work in a number of ways. I estimate the impact of the *MPV* program on a broader set of outcomes, including infant birth weight, the take-up of prenatal care and the number and fraction of births that take place in health facilities (institutions) rather than in the home. Documenting how the *MPV* program affects outcomes beyond fertility is of interest to both policy makers and academics. The former group is interested in the broad impact of the program to properly assess its benefits against the costs. Researchers are also interested in how multifaceted public health interventions like the *MPV* affect outcomes and the mechanisms through which they work. Compared with prior research, I also estimate the potential heterogeneous impact of the program on treated districts that differed in their initial health outcomes and economic activity.

My paper uses aggregated administrative data for the years 2009-2019 (inclusive). The use of administrative health outcome data is beneficial as it does not suffer from imperfect recall and non-responses that are sometimes present in survey data. Furthermore, with multiple years of pre-*MPV* data, I am better able to assess the plausibility of the parallel trends assumption that the DiD methodology relies on. As I discuss below, I find evidence that health outcomes of mothers and infants in treated districts are diverging prior to the introduction of the *MPV* program which motivates my use of the Generalized Synthetic Control methodology.[1].

Finally, I also restrict my analysis to the 2009-2019 period to avoid potential impacts of the COVID-

[1]Although Agarwal et al. (2023) uses three repeated cross sections from the NFHS (1998-99, 2005-06, 2015-16), the intermittent data (surveys every 7 years) makes it difficult to clearly document pre-trends. Furthermore, the 2015-16 wave of the NFHS partially overlaps with the introduction of the *MPV*; as a result, their paper relies only on two years of pre-period data

19 lockdowns in India on the delivery of and response to the MPV program.[2] This is because lockdowns could have affected the accessibility of contraceptives, as noted by Agarwal et al. (2023).

3 Institutional Background

India's child stunting rate is among the highest in the world, exceeding that of many poorer African countries (Jayachandran and Pande, 2017). According to the Annual Report of the Ministry of Health and Family Welfare (Government of India, 2020-21), the major causes of child mortality in India during 2012-13 were prematurity and low birth weight (29.8 %). India's national family planning program has been largely driven by demographic imperatives. Since its inception, the National Health Mission (NHM) has prioritized addressing geographic inequities in fertility and health outcomes and has identified specific districts for targeted investments. Curbing population growth by achieving replacement-level fertility has been the primary focus of the program for over five decades, with female sterilization used as the instrument to deliver replacement-level fertility (Jayachandran and Pande (2017)). While population growth policies target both men and women, historically male sterilization has occurred at lower levels than female sterilization (Salve and Shekhar (2023)).

In November 2016, the Government of India conceived the **'Mission Parivar Vikas'** (MPV) program with a multifaceted approach to substantially increase access to contraceptives and family planning services in 145 high-fertility districts. The districts were part of seven states of India: Assam (2), Bihar (36), Chhattisgarh (2), Jharkhand (9), Madhya Pradesh (25), Rajasthan (14), and Uttar Pradesh (57). Figure 1 is a heat map that shows the spatial variation in fraction of newborns with low birthweight in pre-MPV period - 2009-15 (inclusive). The treated districts are outlined in red and provides visual representation of spatial variation in the outcome of infant health. Treated districts that are shaded darker constitute a larger proportion of newborns with low-brithweight amongst the newborns that were weighed at birth, than the districts in the donor pool[3].

[2] The first COVID-19 lockdown in India began on March 25, 2020.

[3] Heat maps of institutional births, live births and newborns weighed at birth is provided in the Appendix

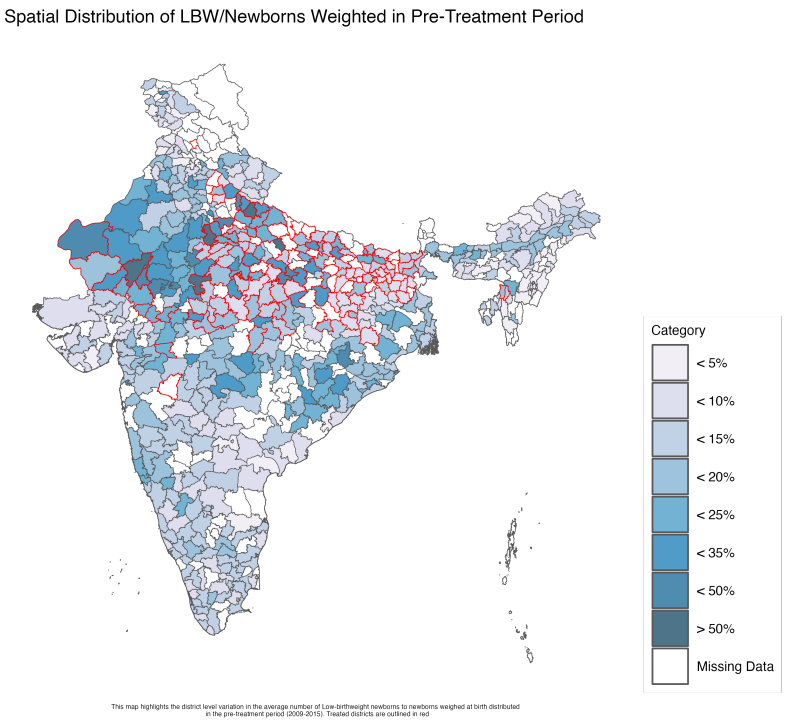


FIGURE 1: This heat map shows the spatial variation of fraction of newborns with low birthweight in India from 2009-15 (inclusive) of treated districts (*outlined in red*) and the donor pool. The darker shade represent a larger share of newborns born with low birthweight amongs the newborns that were weighed at birth.

MPV's unique feature was its multi-pronged approach to reducing fertility. First, the program increased and improved access to a combination of contraceptives such as oral contraceptive pills, condoms, and LARCs from Primary Health Centres (PHCs) and Community Health Centres (CHCs) in both rural and urban areas. The program targeted newlywed couples in rural areas, incentivized social health workers to disseminate family planning kits and information on family planning methods and sent mobile vans equipped with interactive media and family planning products for a fortnight in each quarter of the year.[4] The family planning kits distributed by social health workers include two units of three packeted condoms, two units of combined oral contraceptive pills, three units of emergency contraceptive pills, two units of pregnancy testing kits, and an information card. In rural areas, the *MPV* program also exploited the fact that many couples live in multi-generational households by providing information about contraceptive methods and sexual health among mother-in-laws (MILs) and daughter-in-laws (DILs).[5].

4 Data and Empirical Strategy

4.1 Data

This paper uses district-level aggregate data from the Health Management Information System (HMIS). HMIS is a Government to Government (G2G) web-based monitoring information system that

[4]These vans had a strategic route that covered all *Blocks* within a district.

[5]More details on the policy dissemination of family planning commodities at the district level are in the Appendix. Also included are details of each strategic action proposed in the official Government letter ope-rationalizing the *MPV* program.

has been put in place by the Government of India under the Ministry of Health and Family Welfare (MoHFW) to monitor the National Health Mission, health programs and provide key inputs for policy formulation and appropriate program interventions. The unit of observation is a district and the time period I consider is 2009-2019 (inclusive).

The HMIS was launched in October 2008. Currently, around 225,000 health facilities (across all States/UTs) upload facility-level service delivery data every month, training data every quarter and infrastructure-related data on an annual basis (HMIS, 2024). HMIS coverage is universal.[6] The HMIS contains information on reproductive, maternal and child health-related outcomes, immunization, family planning, vector-borne disease, tuberculosis, morbidity and mortality, OPD (outpatient department), and IPD (inpatient department) services at a monthly frequency for every sub-district in India. For this paper, I focus on birth and maternal outcomes, including total deliveries (live births plus stillbirths), total weighted live births, condoms distributed, stillbirths, the number and fraction of low birthweight babes (i.e. below 2.5 kilgrams or 5.5 pounds), the total pregnant women registering for antenatal care, new cases of hypertension observed in pregnant women registered for antenatal care, institutional births, home-deliveries and the fraction of births where a health worker was present. Since the treatment unit for the *MPV* program is a district, I use the HMIS data aggregated to the district level .

Table 1 reports summary statistics for the pre-period (2009-2015), separately for the treated and control districts. For the dependent variables that are in levels (i.e. total deliveries, total live births, total infants weighed, total low birthweight (LBW) babies, total institutional deliveries, total antenatal registrations and new cases of hypertension), I divide the district level variable by the number of women of reproductive age (15-49) in the district in 2011 using data from the Census of India. This normalization accounts for any baseline differences in the number of women of reproductive age across districts that may affect current and future fertility outcomes. The statistics in Table 1 show that the treated districts experience more births and a higher fraction of LBW babies than the control districts in the pre-MPV period.

In my analysis below, I control for district-level economic activity in some specifications to account for differences in local economies that might affect fertility as well as maternal and child health outcomes. The amount of night lights are a widely used as a proxy for electrification and economic activity when data on local economic outcomes are otherwise unavailable (Henderson et al. (2011); Asher et al. (2021)). Consequently, I use night light data as time-variant covariates for years 2009-19 aggregated at the district level from the Defense Meteorological Program (DMSP) Operational Line-Scan System (OLS) in India. These data provide information on total night lights which are the sum of the luminosity values (0-63) of all pixels in the district and calibrated over time according to the methodology outlined in Ma et al. (2015). In specifications that control for economic activity below, I include the natural logarithm of

[6]For more information on the completeness and accuracy of the HMIS data, see Krishnan et al. (2010), Harikumar et al. (2012), Samal and Dehury (2016).

these total night light values at the district level.

TABLE 1: Summary Statistics

Predictor	**Control**	**Treated**
Newborn Outcomes		
Deliveries	0.0627	0.0836
Live Births	0.0629	0.0834
Weighed	0.0572	0.0726
LBW	0.0079	0.0133
Fraction LBW	0.7550	0.8120
Institutional Deliveries	0.0527	0.0629
Maternal Outcomes		
Antenatal Care Registrations	0.0856	0.1121
New Cases of Hypertension in PW	0.0026	0.0040
Time Variant Covariates		
Night Light (*in ln*)	9.2579	8.9979

Notes: **LBW** = Low Birth Weight infants; **NL** = Night Light; **PW** = Pregnant Women. Night Light data are acquired from the Data Development Lab and Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG).

4.2 Empirical Strategy

The Generalized Synthetic Control Method (GSCM) is used for causal inference using panel data where the unit of observation is a group-time period (e.g. district-year). In this paper I use it to estimate the effect of the *MPV* program on the maternal and child outcomes of treated districts by formulating the counterfactual from the donor pool. The GSCM method estimates the average treatment effect on the treated (ATT) of the *MPV* program by comparing outcomes in the treated districts in the post-2016 period to the outcomes in a synthetic counterfactual district. In this sense, GSCM is similar to the Synthetic Control Method (SCM) proposed by Abadie (2021). Compared to the SCM, the GSCM uses an interactive fixed-effects (IFE) model that interacts unit specific intercepts with time varying coefficients to compute the synthetic counterfactual.

The GSCM has features that make it preferable to other methods such as SCM and DiD in my setting. First, GSCM allows for multiple treated units and different periods. This is especially important in my setting because I have a balanced panel of 126 treated districts. Second, the GSCM estimator proposed by Xu (2017) also includes a method to estimate standard errors, thereby allowing for inference. Third, the GSCM estimator has a built-in cross-validation procedure that automatically selects the correct

number of factors of the IFE model that reduces the risk of over-fitting[7].

Below, I show that a DiD estimator does not exhibit parallel pre-trends in my setting but that the GSCM does. This suggests that the data-driven approach to select comparison units for the counterfactual make it more likely that the counterfactual is plausible. Also, the GSCM method allows for the possibility that the treatment effects of the *MPV* program may be heterogeneous across units (districts).

4.2.1 Framework

Following the notation in Xu (2017), I denote the outcome for unit i un time t by Y_{it}. According to the first assumption of the GSCM—that both the treated and control units are influenced by the same set of underlying factors and that there are no structural breaks during the observed time period—the outcome for unit i at time t can be expressed as follows:

$$Y_{it} = \delta_{it} D_{it} + \beta X_{it} + \lambda_i' f_t + \epsilon_{it} \tag{1}$$

where the treatment indicator D_{it} equals 1 if unit i at time t is exposed to *MPV* treatment and 0 otherwise. Accordingly, δ_{it} is the treatment effect of unit i at time t. X_{it} is a vector of observed covariates, β is a vector of unknown parameters, f_t is a vector of unobserved factors, λ_i is a vector of unknown factor loadings and ϵ_{it} represents unobserved idiosyncratic shocks for the unit i at time t and has a mean of zero. The factor component of the model, $\lambda_i' f_t = \lambda_{i1} f_{1t} + \lambda_{i2} f_{2t} + ... + \lambda_{ir} f_{rt}$, takes a linear and additive form by assumption.

Let $Y_{it}(d)$ denote the outcome of unit i in time t when the treatment indicator is $d \in \{0, 1\}$. The treatement effect of the *MPV* program is given by:

$$\delta_{it} = Y_{it}(1) - Y_{it}(0), t > T_0 \tag{2}$$

Since $Y_{it}(0)$ is not observed for the treated districts, the GSM method estimates the counterfactual using data from the 431 control districts that are in the donor pool. The estimating equation for the counterfactual is:

$$Y_{co} = X_{co}' \beta + F_i' \Lambda_t + \epsilon_{it} \tag{3}$$

where Y_{co} is a (11×431) matrix, X_{co} is a three dimensional matrix (whose dimensions are governed by the number of time periods (11), number of units in the donor pool and number of predictors) and Λ_{co} is a $(431 \times r)$ matrix and $X_{co}'\beta, F\Lambda_{co}', \epsilon_{co}$ are (11×438) matrices. The two constraints that identify

[7]The unobservable factor(s) found by the cross-validation procedure, according to Xu (2017), are not directly interpretable. In a particular case, where only one factor is observed, it might capture a time trend that affects each district differently.

β, F, Λ_{co} are that all factors are normalized and that they are orthogonal to each other. The average difference between the treated unit(s) and its counterfactual in the post-treatment period is given by the following equation:

$$ATT_{t,t>t_0} = \frac{1}{N_{tr}} \sum_{i \in \tau} [Y_{it}(1) - Y_{it}(0)] = \frac{1}{N_{tr}} \sum_{i \in \tau} \delta_{it} \tag{4}$$

Intuitively, I utilize data on health indicators from 2009 to 2019 using HMIS to identify the weighted average of comparison districts that provide the best match for the outcomes. More details about the estimation of the counterfactual are provided in the Online Appendix.

5 Results

I begin by presenting the results from the baseline sample that consists of 126 treated districts and 431 districts in the donor pool.

5.1 Main Results

Estimates of the ATT of the *MPV* program using the GSCM are reported in Table 2. The estimate in column 1 suggests that, on average, the *MPV* program increased deliveries (the sum of live births, still births and infant deaths) by 4.6 infants per one thousand women of reproductive age. The estimates reported in columns 2 and 4 suggest that there is no effect of the *MPV* program on either the number of infants born with low birth weight per 1,000 women of reproductive age or the number of infants born with low birth-weight per 1,000 infants weighed at birth. To contrast, estimates in columns 3 and 5 suggest that the *MPV* program increased the number of infants weighed at birth and the number of births that took place in healthcare facilities by 3.2 and 3.9 infants per one thousand women of reproductive age, respectively. While the *MPV* program was designed to increase access to contraceptives and interaction with the health infrastructure, I estimate small impacts on outcomes such as birth weight of infants that are sensitive to long-term maternal health and nutritional status. In economic terms, the results in Table 2 suggest that the program had a significant impact on outcomes on the extensive margin such as the timing of fertility and the place of birth more than the outcomes on the intensive margin such as maternal nutrition and gestational health.

One of the advantages of the GSCM in this application is using a data-driven approach to select the counterfactual that is more credible than using a DiD estimator. Figure 2 plots the estimates of equation 4 for each year for both the GSCM approach and a dynamic DiD estimator when the dependent variable is total deliveries (i.e. column 1 of Table 2). All estimates reported are relative to the 2015 year (i.e. the year before the introduction of the *MPV* program). In contrast to the GSCM, estimates from the DiD estimator exhibit increasing pre-trends. This suggests that the number of deliveries (relative to the

TABLE 2: ATT ESTIMATES OF HEALTH OUTCOMES OF INFANTS

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
	(1)	(2)	(3)	(4)	(5)
ATT	0.0046***	-0.0006	0.0032**	-0.0015	0.0039**
	(0.0014)	(0.0010)	(0.0016)	(0.0145)	(0.0017)
Unobserved Factors	1	1	1	1	1

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01. There are 126 treated districts with 431 districts in the donor pool.
LBW = Low-Birth Weight infants

number of women of child bearing age) was increasing more quickly in the treated districts than the control districts in the years leading up to the introduction of the *MPV* program. This makes it less plausible that the treated distsricts would have followed the same trends as the control districts in the absence of the policy using the dynamic DiD approach (i.e. the parallel trends assumption). To contrast, Figure 2 shows that the data-driven GSCM selects control districts from the donor pool in a way that leads to parallel pre-trends. The estimates from the GSCM approach also suggest that the treatment effect of the *MPV* program increased each year between 2016-2019.

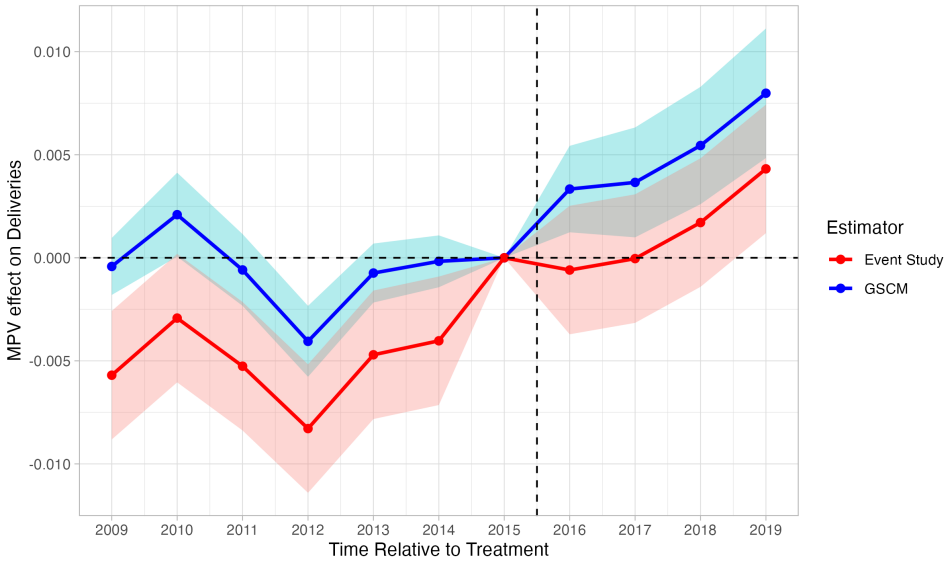


FIGURE 2: Compares the impact of the *MPV* program on health outcome of deliveries per one thousand women of reproductive age in the treated districts using two estimators. The horizontal line (=0) represents the counterfactual. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval. The event study *(in red)* provides evidence that health outcomes of infants in treated districts are diverging prior to the introduction of the *MPV*. This motivates my use of the GSCM *(in blue)*.

5.2 Heterogeneous Treatment Effects

In the preceding section, I reported results for the full sample. In this section, I explore whether the baseline estimates are heterogeneous across districts based on initial health outcomes and economic activity.

5.2.1 *Economic Activity*

Table 3 presents the estimates of the ATT of the *MPV* program for districts based on their level of economic activity in the pre-period period. For brevity, I partition the treated districts into terciles based on their level of night lights in the pre-period. The ATT estimates reported in Table 3 are based on comparisons of the treated districts in each tercile of economic activity/night lights (among the treated units) compared to a counterfactual that is selected from the full group of donor/control districts. Figure 3 plots the estimates of ATT of fraction infants born with LBW for each year for each tercile group. It shows that the *MPV* program increased the fraction of LBW infants in districts in the first tercile initial economic activity. The treatment effect for districts in the second tercile of economic activity is small and statistically insignificant. In contrast, the estimated causal effect of the *MPV* program on the fraction of LBW infants is negative for districts in the top tercile of initial economic activity. In particular, the estimate in column 4 shows that *MPV* program decreased the fraction of infants with low-birthweight by 5.4% in districts with high initial economic activity. To contrast, it increased it in districts with low initial economic activity by 2.8% and no detectable effect on districts with medium initial level of economic activity.

The estimate in column 1 suggest that the *MPV* program increased the number of deliveries for treated districts in the middle tercile of initial economic activity; the estimate suggests that the total number of deliveries increased by 4.2 per one thousand women of reproductive age. However, I estimate that the program had little effect on the number of deliveries in districts in the bottom and top terciles of initial economic activity.

In column 2, the number of infants born with low-birth-weight increased more rapidly in the post-treatment period in districts with low initial economic activity than in districts with higher economic activity. The impact of *MPV* program increased the number of LBW infants by 2 births per one thousand women of reproductive age. The *MPV* program had no significant impact on the number of LBW newborns in districts with higher initial economic activity.

The estimate in column 3 suggests the program also increased the number of infants that were weighed at birth by 8.2 per one thousand women of reproductive age in the districts with relatively high economic activity. It also increased the number of infants weighed at birth by 3.6 per one thousand women in the medium economic activity districts, though this estimate is only statistically significant at the 10 percent level. To contrast, I find no detectable effect on the number of infants weighed at birth in districts with initially low levels of economic activity.

The estimates in column 5 suggest that the *MPV* program increased births in health institutions by 8.8 (resp. 5.2) per one thousand women of reproductive age in the districts with relatively high economic activity (resp. medium economic activity). I estimate that the program had little effect on the number of births in health institutions in the low economic activity districts.

My results are consistent with the *MPV* program having heterogeneous impacts across treated districts with different initial conditions because the counterfactuals for those districts also differed. This implies that the program's effectiveness was not uniform but contingent on local baseline characteristics - in this case district level economic activity. Consequently, districts with lower or higher initial levels of economic activity responded differently to the same intervention, leading to diverging post-*MPV* outcomes. Furthermore, it can be observed from Figures 16 to 19 that districts with higher initial levels of economic activity are more likely to have both sufficient health infrastructure to absorb the program and sufficient unmet fertility demand to respond. However, districts with low initial economic activity lacked implementation capacity (e.g., fewer health centers, lower literacy (Dreze and Murthi (2001)). This aligns with diminishing marginal returns to public health interventions such as the MPV program, wherein its effects sizes are observed to be largest where the constraints are partially but not fully binding. The absence of a measurable impact on low birth weight despite increases in institutional deliveries is consistent with prior evidence that expanding access to maternal services improves care utilization without immediately translating into improved fetal health, which depends more on maternal nutrition, hypertension control, and long-term investments (Powell-Jackson (2015)).

Table 3: ATT ESTIMATES OF HEALTH OUTCOMES OF INFANTS BY INITIAL ECONOMIC ACTIVITY

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
Initial Economic Activity	(1)	(2)	(3)	(4)	(5)
High	-0.0043	-0.0009	0.0082***	-0.0547**	0.0088***
	(0.0036)	(0.0020)	(0.0024)	(0.0274)	(0.0025)
Medium	0.0042**	0.0009	0.0036*	0.0158	0.0052**
	(0.0021)	(0.0009)	(0.0020)	(0.0136)	(0.0024)
Low	0.0006	0.0020***	-0.0030	0.0286***	-0.0023
	(0.0035)	(0.0006)	(0.0037)	(0.0105)	(0.0031)
Unobserved Factors					
High	2	2	1	1	1
Medium	1	1	1	1	1
Low	1	1	2	1	1

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01. There are 42 districts in each treated group with 431 districts in the donor pool that is consistent for each treated group.
LBW = Low-Birth Weight infants; **NL** = Night Light

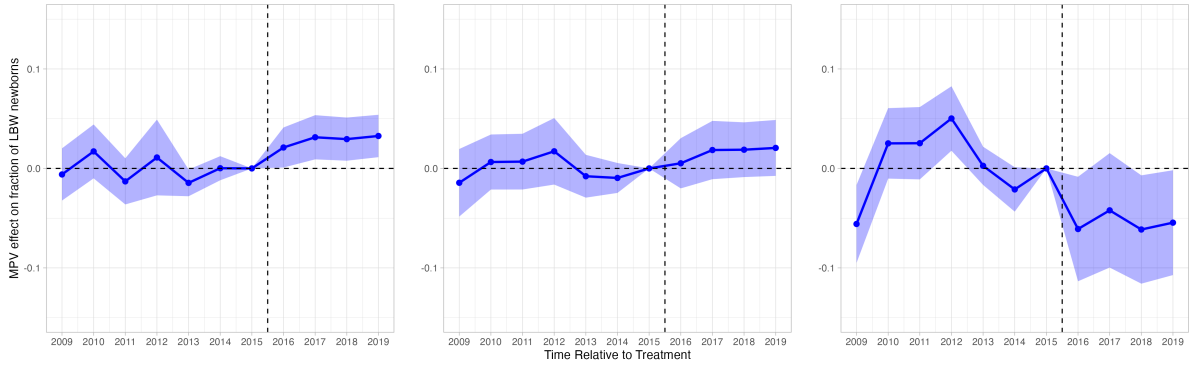


FIGURE 3: From left to right the plots shows the effect of *MPV* program on fraction of infants with low birth-weight (LBW) in districts that differed in their initial economic activity. The figure shows how the treated group's average fraction of infants with low birthweight diverge from the counterfactual over time. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval

5.2.2 *Fraction of Low Birthweight Infants*

Districts in the treatment group have a population equal to 28 percent of the population of India but contribute disproportionately to the negative outcomes experienced by mothers and infants in the country. In particular, the treated districts contribute 30 percent of material deaths, 50 percent of infant deaths and have low modern-contraceptive prevalence rates (Health and Welfare (2016); Jayachandran et al. (2018)). However, one way in which the treated districts differ substantially is in the fraction of low birthweight babies born in the pre-period period. This fraction ranges from 0.05% to around 60% in my sample in the pre-2016 period. As a result of these large initial differences, I also explore the extent to which the *MPV* program has a heterogeneous impact on districts depending on their initial fraction of low birthweight infants. I partition the sample into terciles based on this variable and estimate the impact of the *MPV* program using the GSCM; cutoffs of 12% and 19%. As with the previous section, the donor pool is the full group of untreated districts but the weights assigned to each control district are allowed to vary for each tercile.

Table 4 presents the ATT estimates for each of my main outcomes, separately for each tercile of initial health (fraction LBW). Figure 4 plots the estimates of ATT of fraction infants born with LBW for each year for each tercile. It shows that the *MPV* program increased the fraction of LBW infants in districts with low initial health, with no detectable effect on districts with medium and high initial health levels. Column 4 in Table 4 suggests that the *MPV* program increased fraction of infants with low birthweight by 3.5% in district with low initial fraction of infants with low birthweight. My estimates, in contrary to the results presented in Table 3, suggest that the *MPV* program had no detectable effect on fraction of LBW infants in districts with relatively higher levels of initial health.

Column 1 of Table 4 reports the estimated ATT of the *MPV* program on the number of deliveries per one thousand women of reproductive age in districts with differing initial health levels. My estimates

suggest that the *MPV* program increased the number of deliveries in districts with medium level of initial health by 4.2 newborns. In contrast, the *MPV* program has no detectable effect in districts with high or low initial levels of health.

Column 2 of Table 4 suggests that the *MPV* led to a substantial decrease in the number of newborn born with LBW. In contrast, it increased it in districts with low initial health. On average, the *MPV* program decreased the number of infants per one thousand women of reproductive age in districts with high initial health level by 5.4 infants and increased it in districts with low initial health by 3 infants per one thousand women of reproductive age. My estimates suggest that *MPV* program had no significant effect on districts with medium level of initial health.

Column 3 reports the effect of the *MPV* program on number of newborns weighed at birth per one thousand women of reproductive age. On average, the *MPV* program increased number of infants weighed per one thousand women of reproductive age by 6 and 3.8 in districts with high and medium initial levels of health, respectively. Whereas, districts with low initial level of health, the *MPV* program had no detectable effect on number of newborns weighed at birth.

Column 5 suggests that on average the *MPV* program increased number of institutional births but more so in districts with higher initial levels of health. Specifically, the *MPV* increased births in health institutes by 5.5 and 7.2 per one thousand women of reproductive age in districts with high and medium levels of initial health. My estimates also report that there was no detectable effect of the program on births in health institutes in districts with low initial health. Columns 3 and 5 are consistent, since a newborn is more likely to be weighed if born in a health institute rather than at home.

Table 4: ATT ESTIMATES OF HEALTH OUTCOMES OF INFANTS BY INITIAL HEALTH

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
Initial Health	(1)	(2)	(3)	(4)	(5)
High	-0.0026	-0.0054***	0.0060**	-0.0511	0.0055**
	(0.0025)	(0.0024)	(0.0025)	(0.0342)	(0.0025)
Medium	0.0053**	0.0005	0.0038*	0.0108	0.0072***
	(0.0023)	(0.0007)	(0.0021)	(0.0107)	(0.0025)
Low	0.0016	0.0030***	-0.0005	0.0356***	-0.0010
	(0.0026)	(0.0005)	(0.0031)	(0.0063)	(0.0030)
Unobserved Factors					
High	2	1	1	1	1
Medium	1	1	1	1	1
Low	1	1	2	1	1

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01. There are 42 districts in each treated group with 431 districts in the donor pool that is consistent for each treated group.
LBW = Low-Birth Weight infants

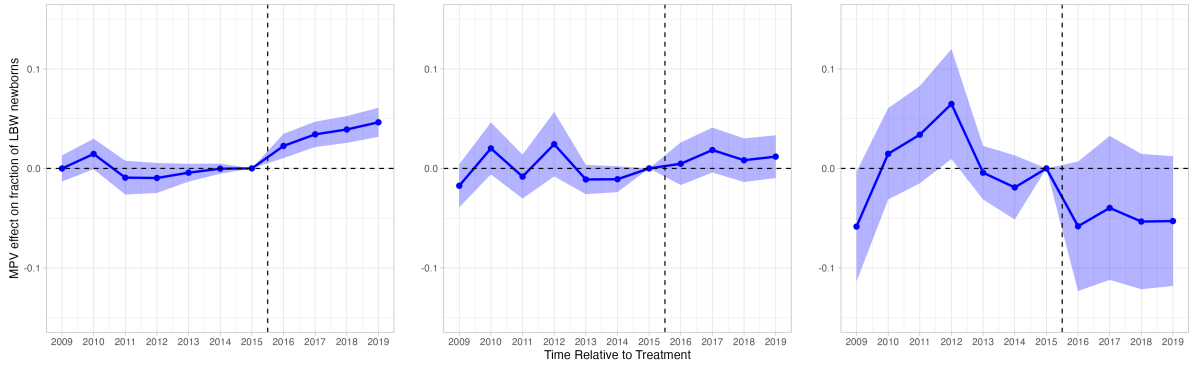


FIGURE 4: From left to right the plots shows the effect of *MPV* program on fraction of infants with low birth-weight (LBW) in districts that differed in their initial health. The figure shows how the treated group's average fraction of infants with low birthweight diverge from the counterfactual over time. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval

5.2.3 *Maternal Outcomes*

In this section, I document the effect of the *MPV* program on maternal health outcomes. This allows me to improve upon the existing evaluations of the program and better understand the impact of the program's overall health implications. Maternal health outcomes offer a window into how integrated public health initiatives, such as the *MPV* program, influence the access to care, engagement with health systems and the well-being of women. Studying maternal outcomes may also lead to insights into the mechanisms through which the *MPV* program works to improve health outcomes mothers and infants.

In this paper, I estimate impact of the *MPV* program on two key maternal health indicators: the uptake of antenatal care services and the incidence of hypertension among pregnant women. These outcomes are critical not only as measures of maternal well-being but also because of their implications for infant health. Hypertension during pregnancy, including conditions such as preeclampsia, poses significant risks to both the mother and the fetus. Clinical research suggests that impaired blood flow resulting from hypertensive disorders can restrict fetal growth, often leading to lower birth weights (Liu et al. (2021); Lim et al. (2014); Wu et al. (2019); Ardissino et al. (2022)).

Table 5 presents the estimates of ATT of the *MPV* program on new cases of hypertension in pregnant women and the number of pregnant women registered for antenatal care. Column 1 indicates that, on average, the *MPV* program resulted in 2 fewer cases of hypertension in pregnant women per one thousand women of reproductive age in all treated districts. Estimates of ATT in districts with high and medium initial economic activity and health suggest, on average, the *MPV* program decreased new cases of hypertension in pregnant women by 1.5 and 3.2 respectively. In contrast, the *MPV* program had no effect on new cases of hypertension in pregnant women in treated districts with low initial economic activity and health.

Column 2 of Table 5 suggests that the *MPV* increased 21.2 and 12.9 more pregnant women to register

for antenatal care in districts with high and low initial health and economic activity. Figure 5 and 6 show the treated group's ATT coefficients diverge from the counterfactual over time for new cases of pregnant women with hypertension and the take-up of antenatal care. Specifically, districts that exhibit decrease in the fraction of newborns with LBW shown in Figure 3 also exhibit decreasing incidences of hypertension found in pregnant women shown in Figure 5.

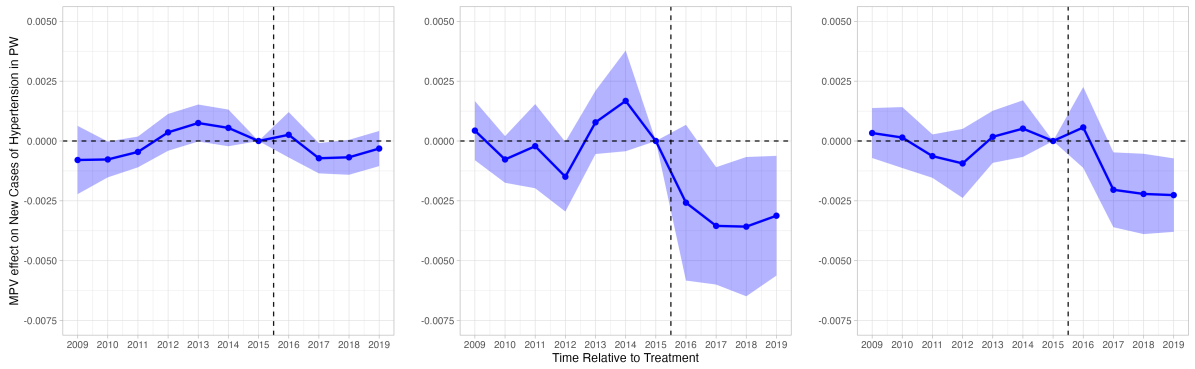


FIGURE 5: From left to right the plots shows the effect of MPV program on new cases of hypertension found in pregnant women in districts that differed in their initial health. The figure shows how the treated group's average number of pregnant women with hypertension diverge from the counterfactual over time. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval

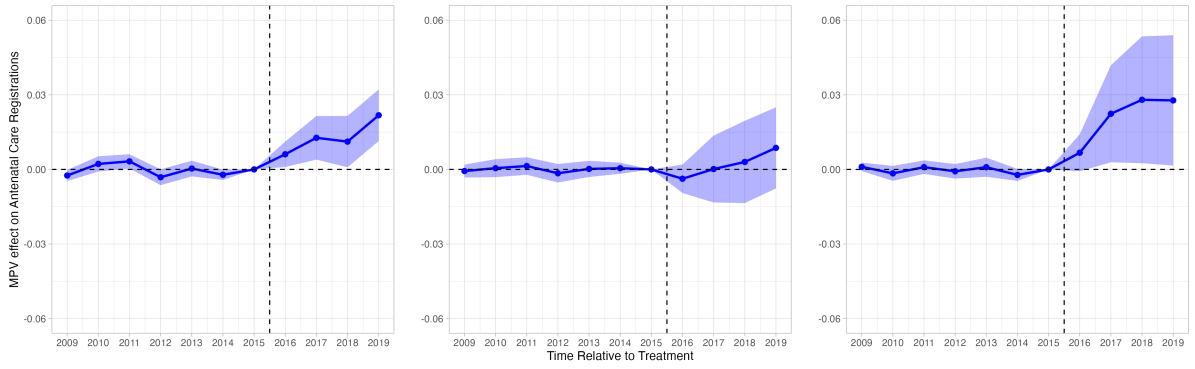


FIGURE 6: From left to right the plots shows the effect of MPV program on number of antenatal care registrations of pregnant women in districts that differed in their initial health. The figure shows how the treated group's average number of pregnant women registered for antenatal care diverge from the counterfactual over time. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval

6 Conclusion

This paper investigates the impact of the understudied MPV program launched in India in 2016, which expanded access to contraceptives and family planning services across high-fertility districts in seven states. Family planning programs have long been central to public health efforts, particularly in low- and middle-income countries, where high fertility rates and constrained access to reproductive health services can limit economic opportunities and adversely affect maternal and child health. A large

Table 5: ATT ESTIMATES OF HEALTH OUTCOMES OF MOTHERS BY INITIAL HEALTH AND ECONOMIC ACTIVITY

Dependent Variable	New Cases of Hypertension	Antenatal Care Registrations
	(1)	(2)
	-0.0019***	0.0055
	(0.0006)	(0.0034)
Initial Health		
High	-0.0015**	0.0212***
	(0.0007)	(0.0077)
Medium	-0.0032***	0.0020
	(0.0004)	(0.0050)
Low	-0.0004	0.0129***
	(0.0003)	(0.0032)
Initial Economic Activity		
High	-0.0015***	0.0212***
	(0.0004)	(0.0067)
Medium	-0.0032**	0.0020
	(0.0004)	(0.0035)
Low	0.0004	0.0129***
	(0.0004)	(0.0031)
Unobserved Factors		
	1	2
High	1	3
Medium	1	2
Low	0	2

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.
LBW = Low-Birth Weight infants; **NL** = Night Light.
Treated and control districts in new cases of hypertension in pregnant women are taken for all districts that report greater than 0 cases in the post treatment period 20016-2019 (inclusive). Donor pool consists of 350 and 431 districts for new cases of hypertension and antenatal care registrations respectively.

body of literature has evaluated the demographic and fertility-related consequences of family planning policies, relatively less attention has been paid to their downstream effects on maternal and infant health outcomes. By focusing on the *MPV* program, this study contributes to a growing empirical literature that examines how improved access to reproductive health services can influence the quality of care during pregnancy and childbirth.

This study builds on existing research by estimating the impact of the *MPV* program on a broader set of health outcomes of both infants and mothers, leveraging the Generalized Synthetic Control Method (GSCM) proposed by Xu (2017). This method employs a data-driven procedure to construct a counterfactual by optimally weighting districts in the donor pool to closely match pre-period outcomes of the treated districts. I use administrative data covering the years 2009 to 2019 (inclusive) for the analysis. Finally, study also explores the potential heterogeneity in program effects based on initial levels of health and economic activity across districts.

The results indicate that, on average, the *MPV* program increased the number of deliveries by 4.6 per one thousand women of reproductive age in treated districts. Among these, 3.9 live births occurred in health institutions, and 3.2 of the infants per one thousand women were weighed at birth. In addition, the program is associated with a reduction of 1.9 new cases of hypertension per one thousand pregnant women. Estimates of heterogeneous treatment effects reveal that in districts with higher initial economic activity, the *MPV* program reduced the share of infants with low birth weight by 5.4% and reduced the incidence of hypertension in pregnant women by 1.5 cases per one thousand. In contrast, in districts with low initial economic activity, the program increased the prevalence of low-birth-weight infants by 2.8% and had no discernible effect on the incidence of maternal hypertension. These findings underscore the importance of contextual factors—such as local economic capacity—in shaping the effectiveness of large-scale family planning interventions.

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A Appendix

A.1 Formulation of Counterfactual

The general idea of GSCM is to choose the model that leads to the most accurate predictions. To be able to find the model, some of the data is withheld and the remaining part of the data is used to make predictions for the withheld part and the model then provides the most accurate predictions is the chosen model (with requirement that the mean squared predicted error is minimized). The first step is to obtain a fixed number of latent factors by estimating an IFE model using the data of the donor pool (control group):

$$(\hat{\beta}, \hat{F}, \hat{\Lambda}_{co}) = \min_{\hat{\beta},\hat{F},\hat{\Lambda}_{co}} \sum_{i\in\tau} (\tilde{Y}_i - \tilde{X}_i\beta - \tilde{F}\tilde{\lambda}_i)'(\tilde{Y}_i - \tilde{X}_i\beta - \tilde{F}\tilde{\lambda}_i)$$

$$s.t. \frac{\tilde{F}'\tilde{F}}{T} = I_r \; and \; (\tilde{\Lambda}'\tilde{\Lambda})$$

The second step is to estimate the factor loadings $\hat{\lambda_i}$ (unit specific intercepts) by projecting pre-treatment outcomes of the reated unit(s) on the space spanned by the factors estimated in the first step.

$$\begin{aligned}\hat{\lambda}_i &= \min_{\hat{\Lambda}_i}(Y_i^0 - X_i^0\hat{\beta} - \hat{F}^0\tilde{\lambda}_i)'(Y_i^0 - X_i^0\hat{\beta} - \hat{F}^0\tilde{\lambda}_i) \\ &= (\hat{F}^{0\prime}\hat{F}^0)^{-1}\hat{F}^{0\prime}(\hat{Y}^0 - \hat{X}^0\hat{\beta}^0), \; i \in \tau,\end{aligned}$$

The third step is to estimate the counterfactual based on the factors and factor loadings computed in steps one and two by:

$$\hat{Y}_{it}^0 = x_{it}'\hat{\beta} + \hat{\lambda_i}'\hat{f}_t, \;\; i \in \tau, t > T_0$$

A.2 Robustness Checks

I conclude the empirical analysis of the MPV program by assessing the robustness of the main results, focusing on the treated and non-treated districts and their trajectories in treated states.

One way to check the robustness of these results is to compare the outcomes of the treated districts within each state with the districts that were not part of the *MPV* program in that respective state. This comparison utilizes the districts that are most similar to the treated districts as they are within the state and pre-period trends would be more liable to meet parallel trends. Table 6 shows the ATTs of the *MPV* program on the health outcomes of treated districts in treated provinces with the donor pool restricted to only the districts that were not included in the *MPV* program within the respective state. Table 6 shows that when the treated districts in the treated states are compared to non-treated districts within that state, on average, has not had significant effect on the health outcomes of mothers and infants[8]. Furthermore, in the state of Uttar Pradesh the number of infants with low brithweight per 1000 women of reproductive age on average increased by 167, which is quite large. However, the donor pool is smaller than the treated group of 49 in this state.

[8]The treated state of Bihar is omitted in this robustness check as a majority of districts in this state were included in the *MPV* program. Thus, making the donor pool for this robustness check unsuitable to take the weighted average of only two districts to compute the counterfactual of 38 treated districts in the state of Bihar.

TABLE 6: ATT ESTIMATES OF HEALTH OUTCOMES OF MOTHERS AND INFANTS IN TREATED DISTRICTS WITH DONOR POOLS RESTRICTED TO CONTROL DISTRICTS IN TREATED STATES.

State	AS	CH	JR	UP	RJ	MP
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Deliveries	0.011	-0.084	0.015	0.037	-0.039	-0.007
	(0.062)	(0.275)	(0.052)	(0.08)	(0.05)	(0.023)
LBW	0.236	-0.161	0.088	0.167**	-0.082	0.075
	(0.266)	(0.324)	(0.124)	(0.082)	(0.085)	(0.104)
Live Births	0.015	0.109	0.059	0.087	-0.016	-0.012
	(0.062)	(0.092)	(0.053)	(0.053)	(0.042)	(0.023)
Infants Weighted	-0.011	0.11	0.011	-0.163	-0.021	-0.012
	(0.099)	(0.114)	(0.038)	(0.156)	(0.038)	(0.022)
Fraction LBW	0.000	0.012	0.007	0.016	-0.029	0.015
	(0.018)	(0.025)	(0.012)	(0.043)	(0.020)	(0.018)
Fraction of Infants Weighed	-0.031	-0.017	-0.005	0.024	-0.023	0.002
	(0.064)	(0.029)	(0.015)	(0.044)	(0.020)	(0.011)
Antenatal Care Registrations	0.02	0.056	0.015	-0.034	0.043	-0.072
	(0.073)	(0.115)	(0.036)	(0.057)	(0.029)	(0.046)
Institutional Births	0.032	-0.119	0.069	-0.091	-0.02	0.001
	(0.092)	(0.284)	(0.064)	(0.089)	(0.044)	(0.032)
Still Births	-0.057	-0.622	0.210	0.206	0.009	0.050
	(0.258)	(0.488)	(0.262)	(0.159)	(0.17)	(0.07)
Treated Districts	2	2	9	49	13	25
Control Districts	25	12	15	13	19	24

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.

LBW = Low-Birth Weight infants; **NL** = Night Light

AS = Assam; **CH** = Chattisgarh; **JR** = Jharkhand; **UP** = Uttar Pradesh; **RJ** = Rajasthan; **MP** = Madhya Pradesh.

A further robustness check is to pool the untreated districts within the treated states and compare their health outcomes to those of the treated districts. This serves two purposes. First, it tests the validity of the Generalized Synthetic Control Method (GSCM); the consistency of results under this restricted control group supports the credibility of the initial findings. Second, it provides a test for potential spillover effects. If substantial spillovers were present, I would expect the untreated districts in the same states to exhibit similar outcome trajectories to the treated districts. The persistence of the treatment effect despite this restriction suggests that spillovers are limited.

Table 7 presents the ATTs of the treated districts with the donor pool restricted to the pooled districts that were not part of the MPV program but are in the seven provinces that were treated by the MPV program. Although, the signs of low-birthweight infants, fraction of low-birthweight infants, instituional deliveries and antenatal care registrations per thousand women of reproductive age are

reverse, they are of smaller magnitude and I do not detect any significant effect of the MPV program on the treated districts. This suggests large spillover effects in districts that were not part of the MPV program but geographically located in the provinces that were treated.

TABLE 7: ATT ESTIMATES OF HEALTH OUTCOMES OF TREATED DISTRICTS WITH THE DONOR POOL RESTRICTED TO THE POOLED CONTROL DISTRICTS IN TREATED STATES

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries	Antenatal Care Registrations
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	0.0016	0.0016	0.0022	0.0314	-0.0002	-0.0007
	(0.0019)	(0.0014)	(0.0017)	(0.0224)	(0.0019)	(0.0036)
Unobserved Factors	1	1	1	1	1	2

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01. There are 126 treated districts with 108 districts in the donor pool.
LBW = Low-Birth Weight infants

A.3 ATT Estimates

The following section proceeds to present the figures for treated group's ATT coefficients diverge from the counterfactual over time for infant health outcomes. The gap plots are representative of the ATT presented in Table 2.

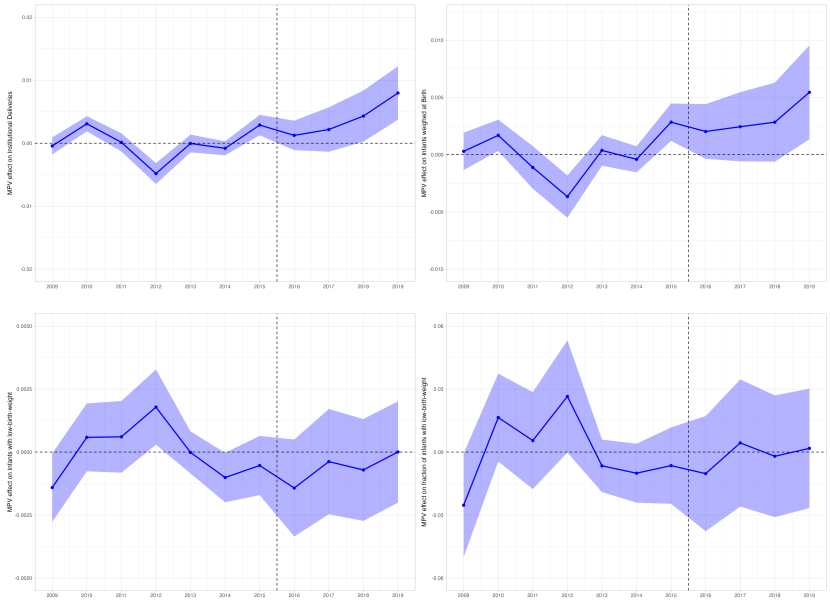


FIGURE 7: The figure shows how the treated group's average outcome diverge from the counterfactual over time. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

A.3.1 ATT Estimates by Initial Economic Activity

The following section proceeds to present the figures for treated group's ATT coefficients diverge from the counterfactual over time based on their pre-treatment initial health levels. The gap plots are representative of the ATT estimated presented in Table 3 for outcome of infants that are in levels:

deliveries, institutional deliveries, infants weighed at birth and number of infants born with low birth-weight per one thousand women of reproductive age.

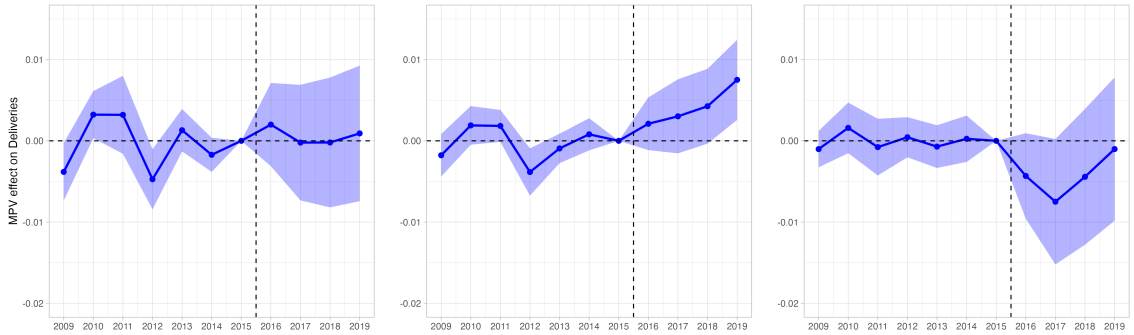


FIGURE 8: The figure shows how the treated group's average number of deliveries per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of economic activity. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

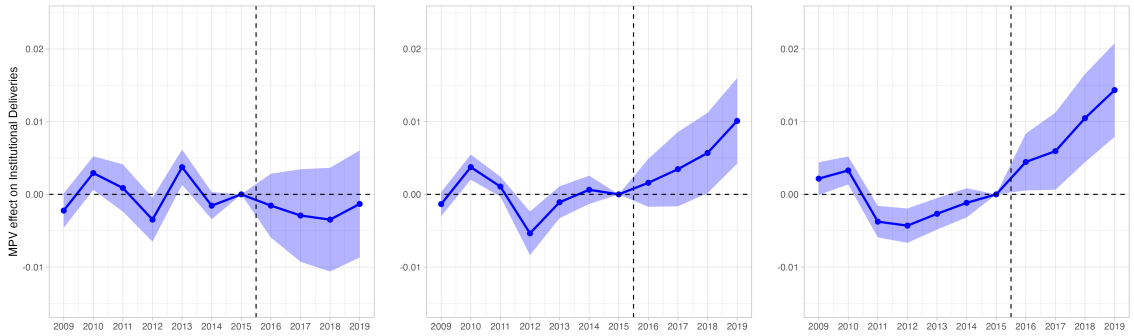


FIGURE 9: The figure shows how the treated group's average number of institutional deliveries per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of economic activity. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

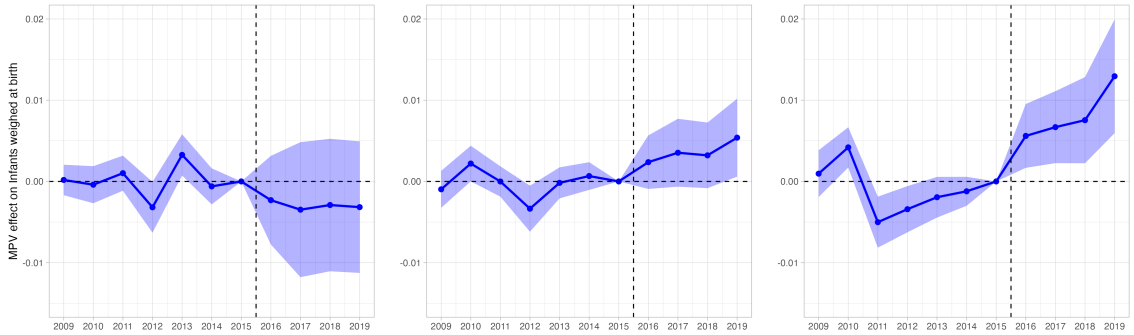


FIGURE 10: The figure shows how the treated group's average number of infant weighed at birth per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of economic activity. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

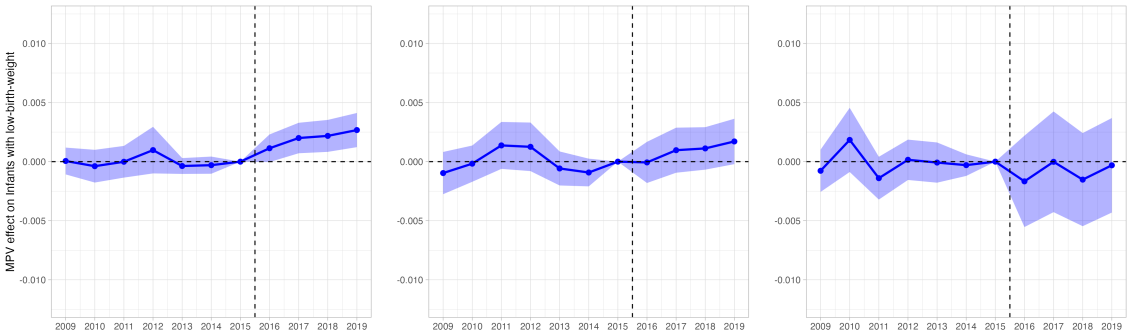


FIGURE 11: The figure shows how the treated group's average number of infant with low birth weight per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of economic activity. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

A.3.2 ATT Estimates by Initial Health

The following section proceeds to present the figures for treated group's ATT coefficients diverge from the counterfactual over time for treated districts based on their pre-treatment initial health levels. The gap plots are representative of the ATT estimated presented in Table 4 for outcome of infants that are in levels: deliveries, institutional deliveries, infants weighed at birth and number of infants born with low birth-weight per one thousand women of reproductive age.

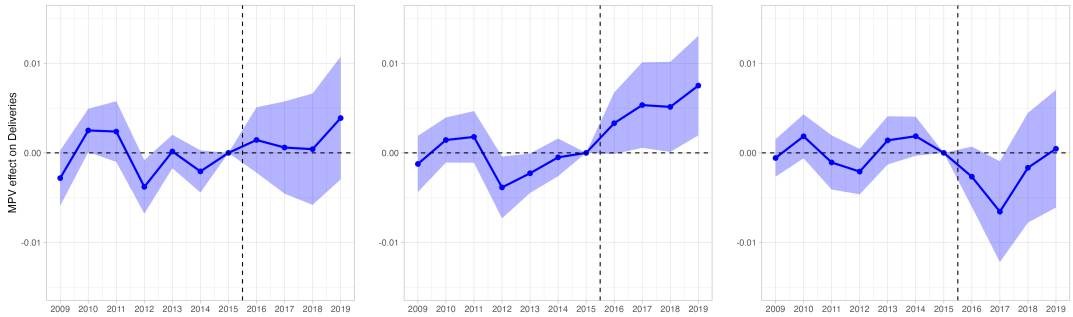


FIGURE 12: The figure shows how the treated group's average number of deliveries per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of health. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

TABLE 8: AVERAGE FUNCTIONAL HEALTH CENTERS

	Sub centres	PHCs	CHCs	Div.Hosp	District Hosp
Control	219.51	33.54	8.10	1.82	1.05
High IEA	291.95	41.86	11.32	0.56	1.62
Med IEA	271.49	40.51	7.55	1.03	1.42
Low IEA	188.83	27.34	4.71	0.49	1.04
High IH	278.88	42.10	10.13	0.30	1.70
Med IH	246.76	34.56	7.65	0.83	1.28
Low IH	227.13	33.17	5.92	0.92	1.11

CHC: Community Health Centers; PHC: Public Health Centers; Dist Hospital, Sub-centers, and Sub-divisional hospitals IEA: Initial Economic Activity; IH: Initial Health

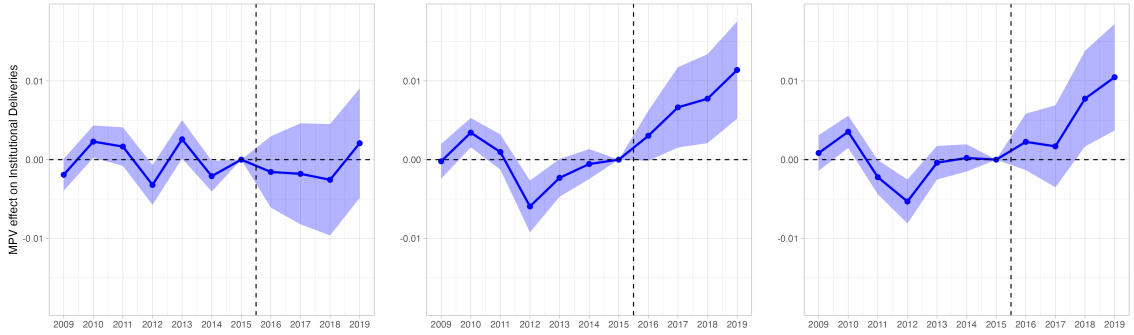


FIGURE 13: The figure shows how the treated group's average number of institutional deliveries per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of health. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

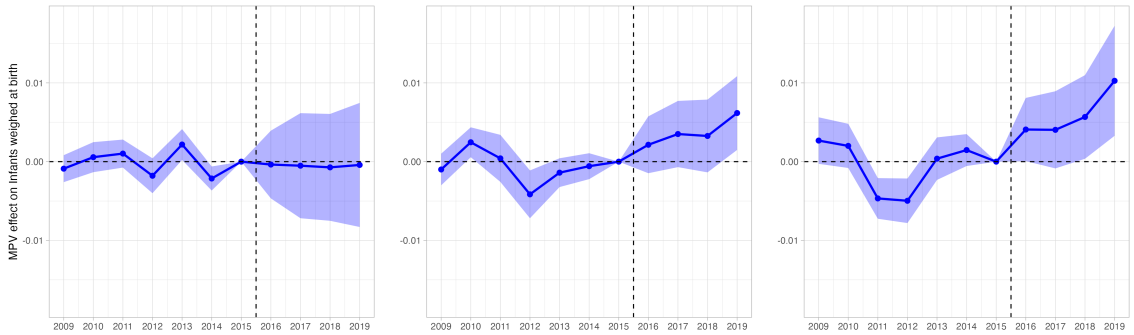


FIGURE 14: The figure shows how the treated group's average number of infant weighed at birth per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of health. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

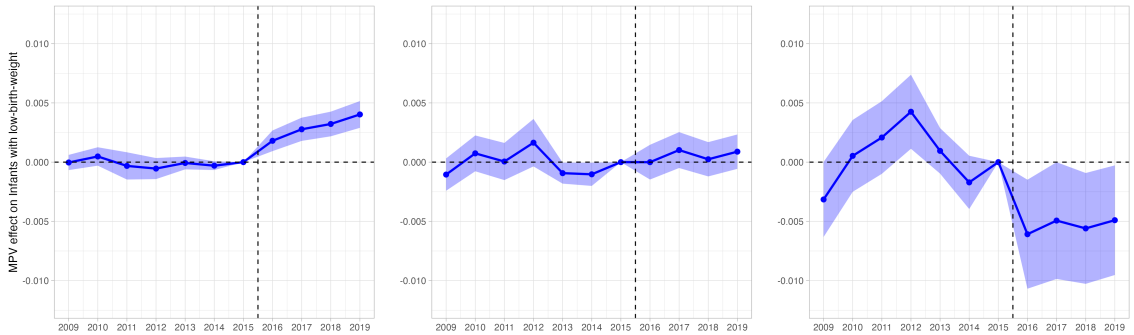


FIGURE 15: The figure shows how the treated group's average number of infant with low birth weight per one thousand women of reproductive age diverge from the counterfactual over time. From left to right the treated districts represent low, medium and high initial levels of health. The horizontal line (=0) represents the counterfactual and the blue line represents the the outcome of the treated group. The vertical line (=0) is the treatment year. The shaded area is the 95% confidence interval.

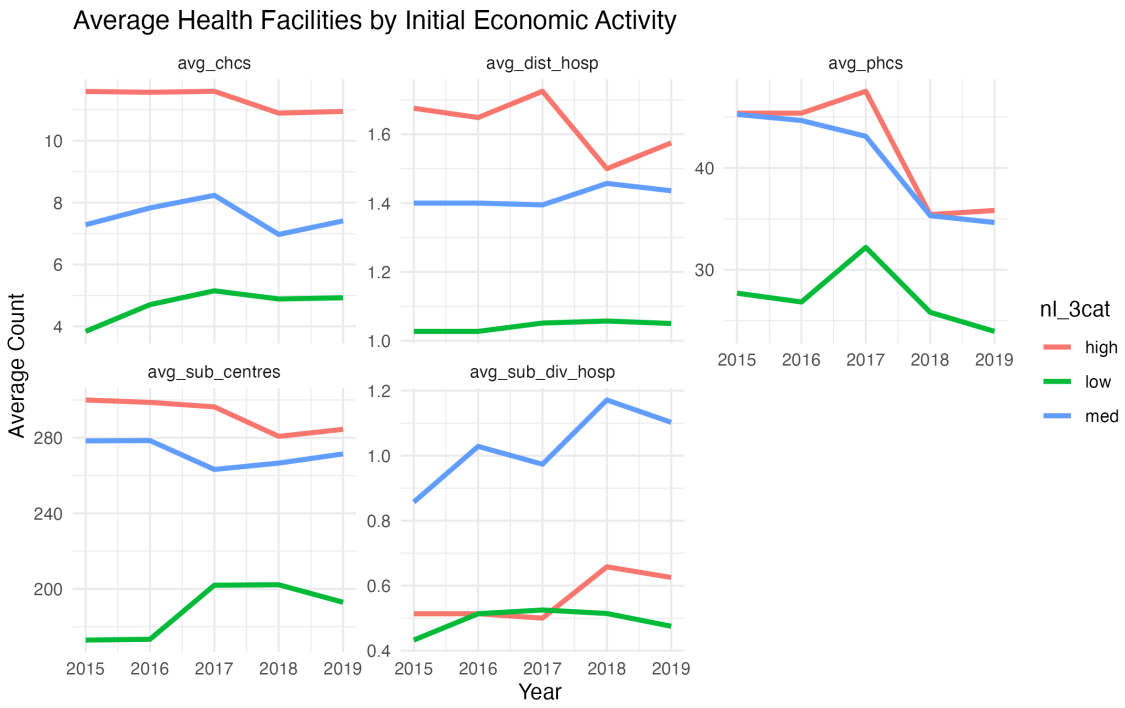


FIGURE 16: The figure shows the trends of different types of functional health centers (CHC: Community Health Centers; PHC: Public Health Centers; Dist Hospital, Sub-centers, and Sub-divisional hospitals) of treated districts for periods: 2015 to 2019 by their initial economic activity.

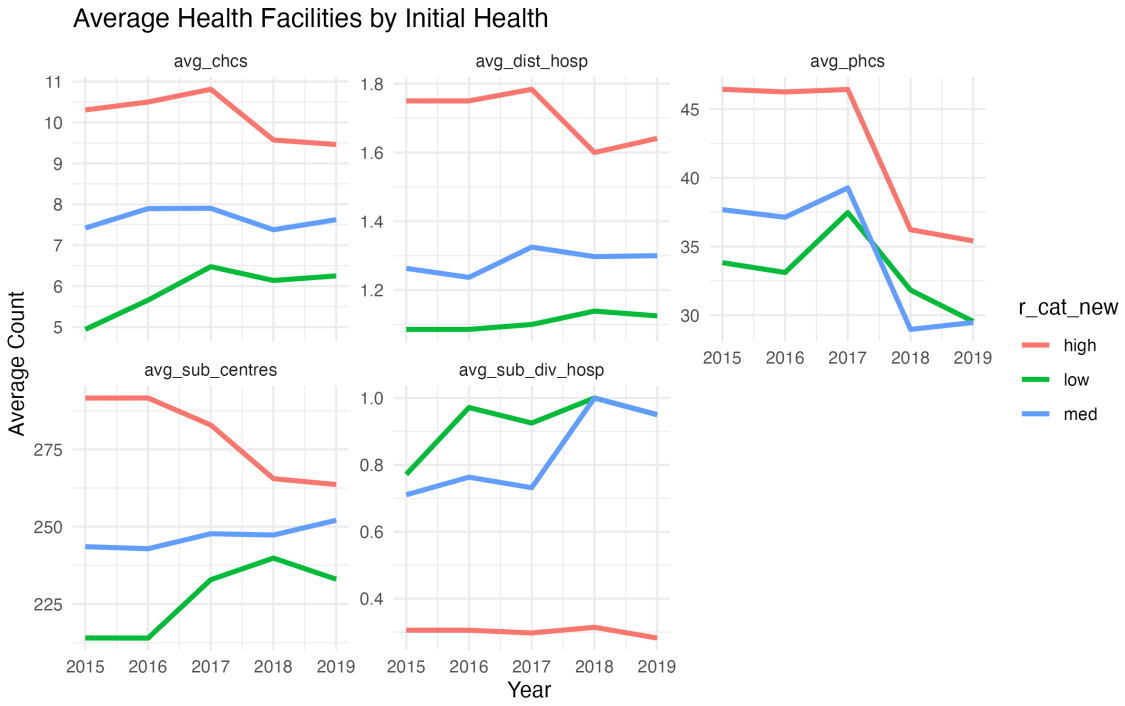


FIGURE 17: The figure shows the trends of different types of functional health centers (CHC: Community Health Centers; PHC: Public Health Centers; Dist Hospital, Sub-centers, and Sub-divisional hospitals) of treated districts for periods: 2015 to 2019 by their initial health.

TABLE 9: FUNCTIONAL HEALTH CENTERS PER THOUSAND WOMEN FOR PERIOD 2015-2019

	Sub centres	PHCs	CHCs	Div.Hosp	District Hosp
Control	0.00032	0.00005	0.00001	0.000003	0.000002
High IEA	0.00331	0.00047	0.00013	0.00001	0.00002
Med IEA	0.00342	0.00051	0.00010	0.00001	0.00002
Low IEA	0.00335	0.00049	0.00008	0.00001	0.00002
High IH	0.00358	0.00054	0.00013	0.000004	0.00002
Med IH	0.00328	0.00046	0.00010	0.00001	0.00002
Low IH	0.00321	0.00047	0.00008	0.00001	0.00002

Sub centres, PHCs, CHCs, Div.Hosp, and District Hosp indicate the number of health facilities per 1000 women in each category.

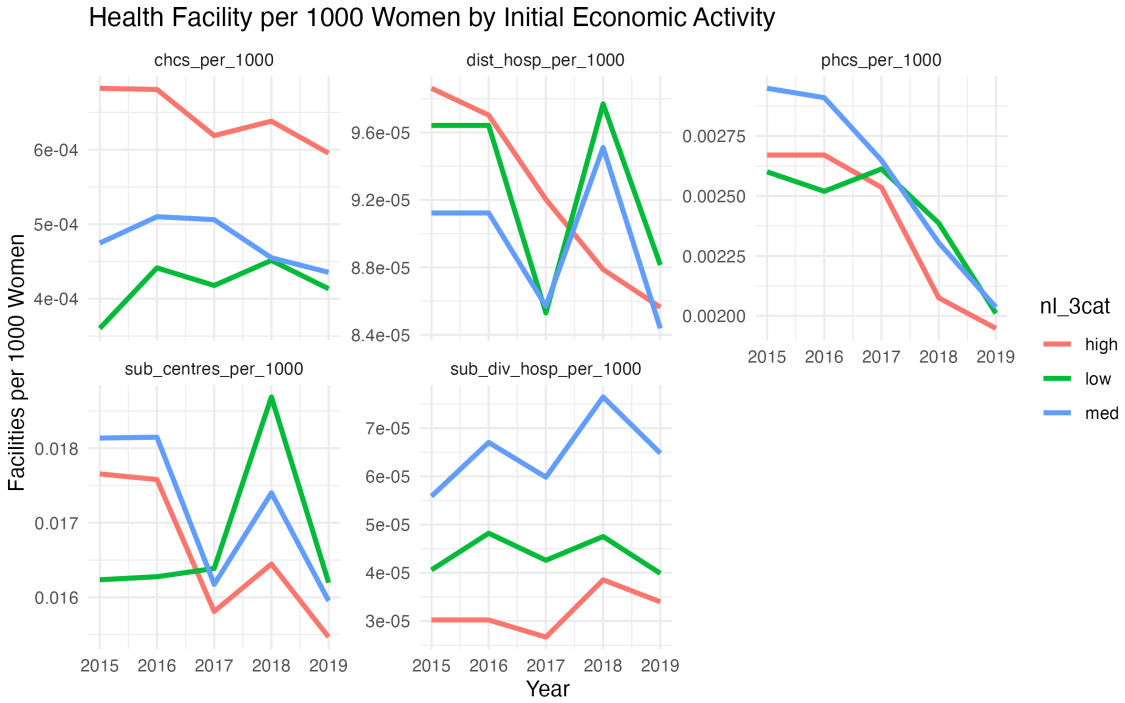


FIGURE 18: The figure shows the trends of different types of functional health centers by 1000 women ever married women (CHC: Community Health Centers; PHC: Public Health Centers; Dist Hospital, Sub-centers, and Sub-divisional hospitals) of treated districts for periods: 2015 to 2019 by their initial economic activity.

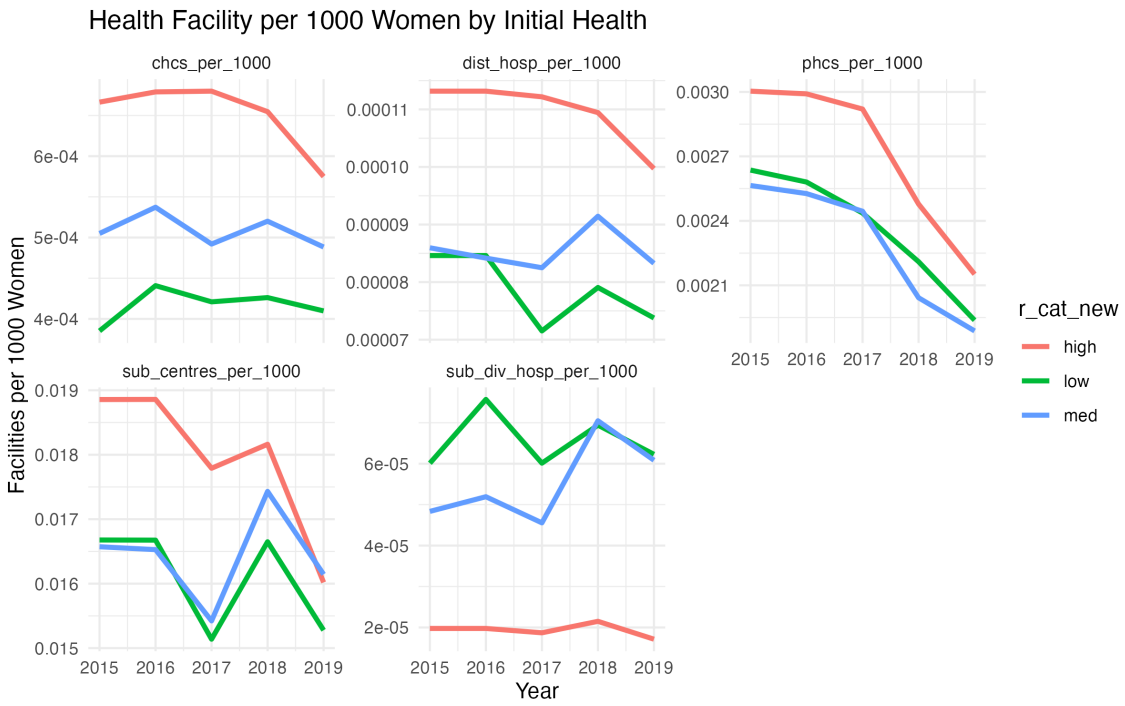


FIGURE 19: The figure shows the trends of different types of functional health centers by 1000 women of reproductive age (CHC: Community Health Centers; PHC: Public Health Centers; Dist Hospital, Sub-centers, and Sub-divisional hospitals) of treated districts for periods: 2015 to 2019 by their initial health.

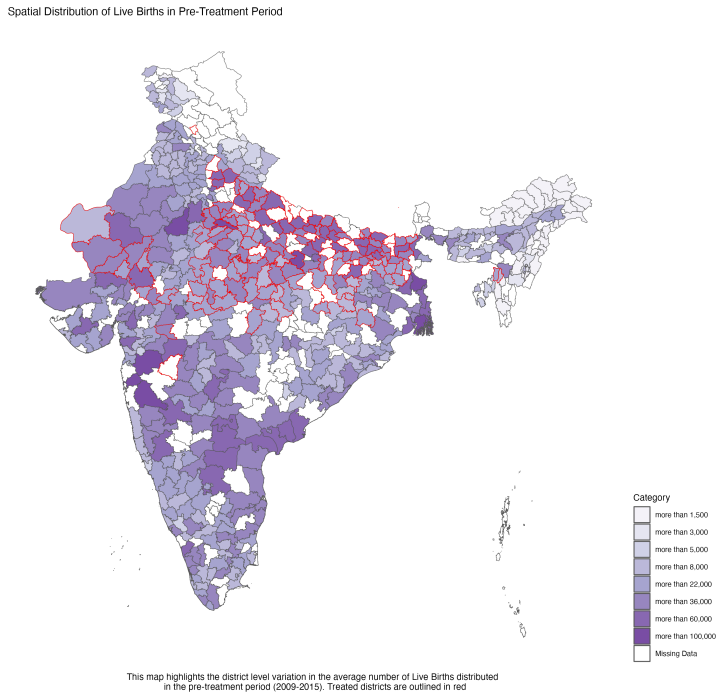


FIGURE 20: This heat map shows the spatial variation of live births in India from 2009-15 (inclusive) of treated districts (*outlined in red*) and the donor pool.

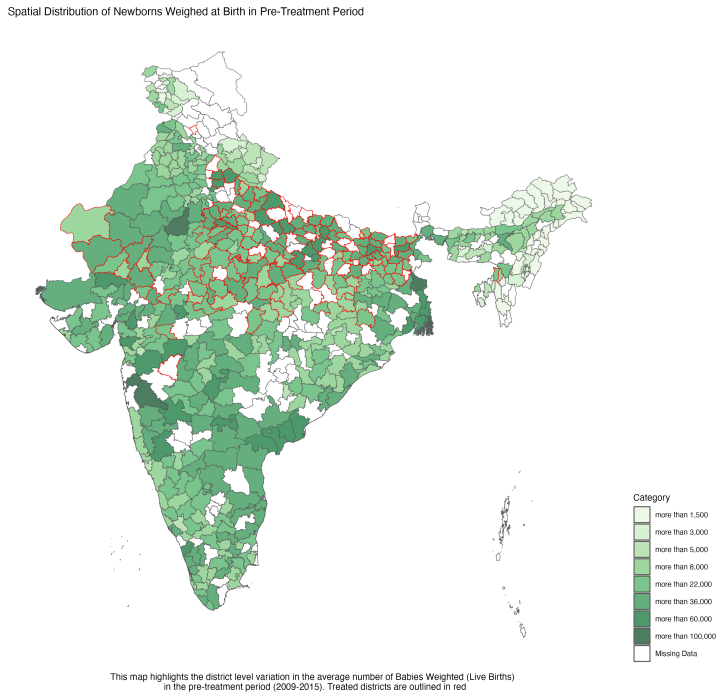


FIGURE 21: This heat map shows the spatial variation of live newborns that were weighed at birth in India from 2009-15 (inclusive) of treated districts (*outlined in red*) and the donor pool.

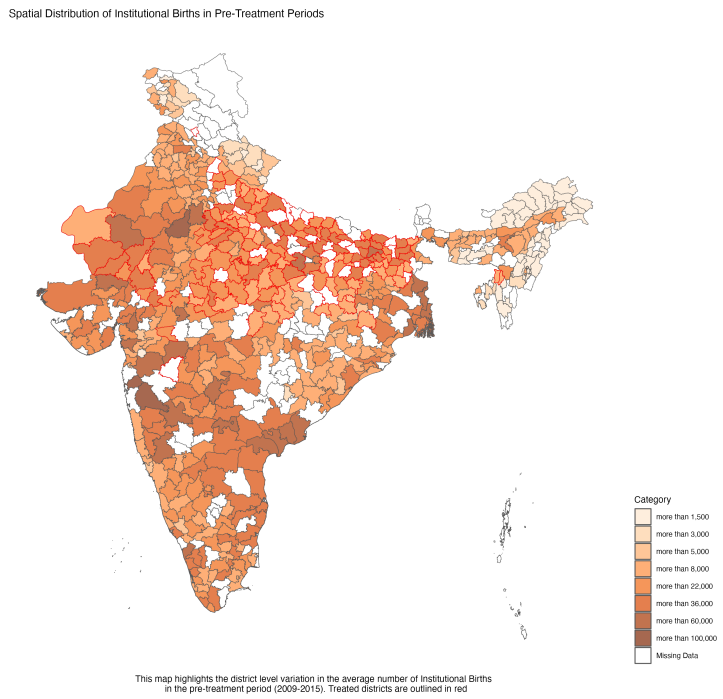


FIGURE 22: This heat map shows the spatial variation of institutional births in India from 2009-15 (inclusive) of treated districts (*outlined in red*) and the donor pool.

B Conflict of Interest

The authors declare that they have no conflict of interest.

C Data Access, Repository and Acknowledgment

Any opinions and conclusions expressed herein are those of the author and do not represent the views of the Ministry of Health and Family Welfare, Government of India. Data are available to the public on the Health Management Information System (HMIS) website https://hmis.mohfw.gov.in.